Skill Intelligence Dashboard

Extracting and Visualizing Skills from Activity Logs

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Coriolis Internship Assignment Presentation

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Problem Statement

Why do we need *Skill Intelligence*?

Modern teams generate large amounts of activity logs, task descriptions, and project notes.
However:

- Skills are *hidden inside unstructured text*
- It is difficult to measure *team expertise*
- Managers lack *data-driven insights* about skill distribution
- Manual analysis of logs is *time-consuming and inaccurate*

Goal: Automatically extract and visualize skill patterns.

Project Goal

The goal of this project is to build a system that:

- Extracts technical skills from activity logs
- Groups them into hierarchical skill categories
- Generates interactive visual analytics
- Helps understand *skill distribution and development patterns*

This transforms *unstructured activity logs into structured skill intelligence.*

System Overview

The system processes activity logs and converts them into interactive analytics.

Pipeline:

Dataset Upload
↓
Skill Extraction
↓
Skill Aggregation
↓
Interactive Dashboard

Users can upload *CSV or Excel datasets*, and the system automatically analyzes skill usage.

System Architecture

The dashboard is built using a modular architecture:

Frontend Layer

- Streamlit dashboard

Processing Layer

- Skill extraction engine

Aggregation Layer

- Skill grouping system
- Visualization Layer

Plotly charts

- Network graphs

This separation allows the system to remain *scalable and maintainable*

Skill Extraction Engine

The system scans activity logs and detects skill mentions.

Steps:

1. Load dataset
2. Extract text fields (tasks, notes, project titles)
3. Match phrases using predefined skill dictionary
4. Count skill frequencies

Example detected skills:

- GPT
- Quantization
- LoRA
- Finetuning

Extraction is implemented using *regex-based matching with optional NLP support*

Hierarchical Skill Aggregation

Extracted skills are grouped into **parent categories**.

Example hierarchy:

- Large Language Models
  - GPT
  - BERT
  - ChatGPT

Model Optimization

- Quantization
- LoRA
- FP16

Low-frequency skills are automatically merged into their parent categories using a *threshold rule*. This simplifies visualization and improves interpretability.

Dashboard Features

The system provides multiple analytics views:

- KPI Metrics
- Skill Frequency Distribution
- Skill Hierarchy Treemap
- Skill Relationship Network
- AI skill clustering
- Multi-User Comparsion
- Skill Evolution Timeline

All charts are generated using *Plotly and NetworkX* for interactive exploration

Demo Results

The dashboard allows users to:

- Identify the most frequent skills
- Analyze skill relationships
- Compare skill distribution across users
- Track skill development trends

Example insight:

LLM-related skills dominate activity logs, indicating strong focus on AI systems.

Conclusion

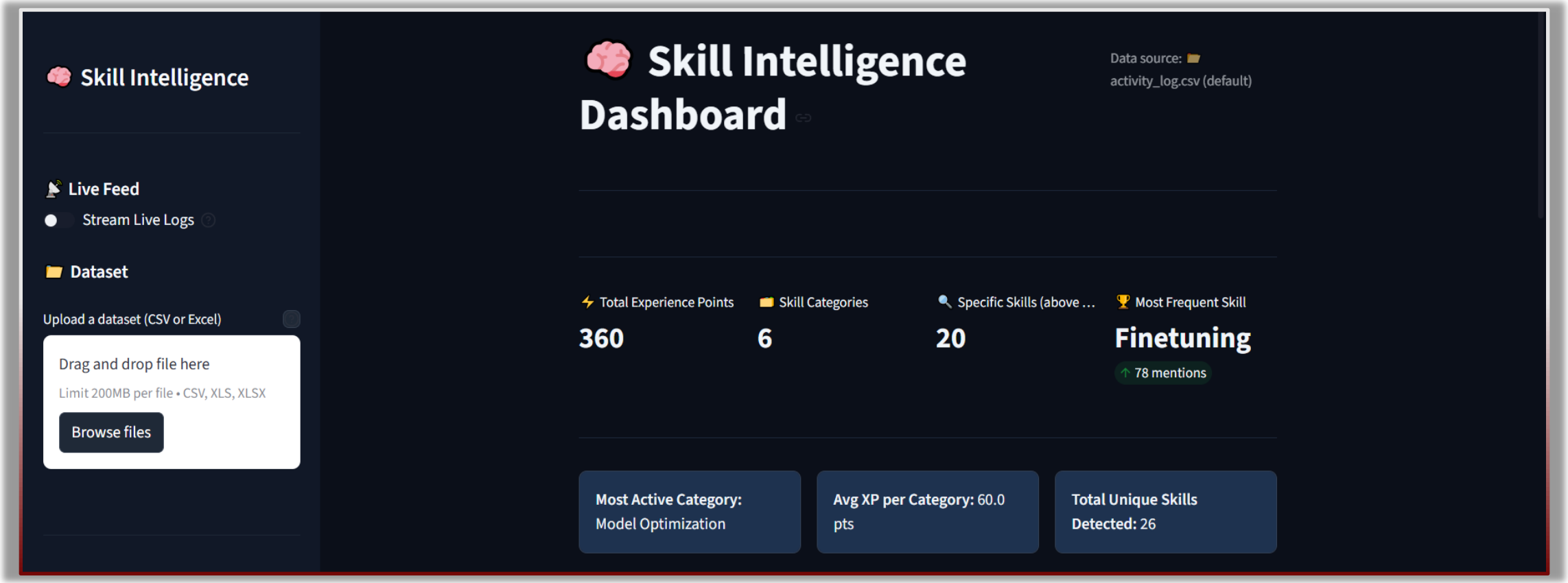
This project demonstrates how activity logs can be transformed into structured insights.

Key contributions:

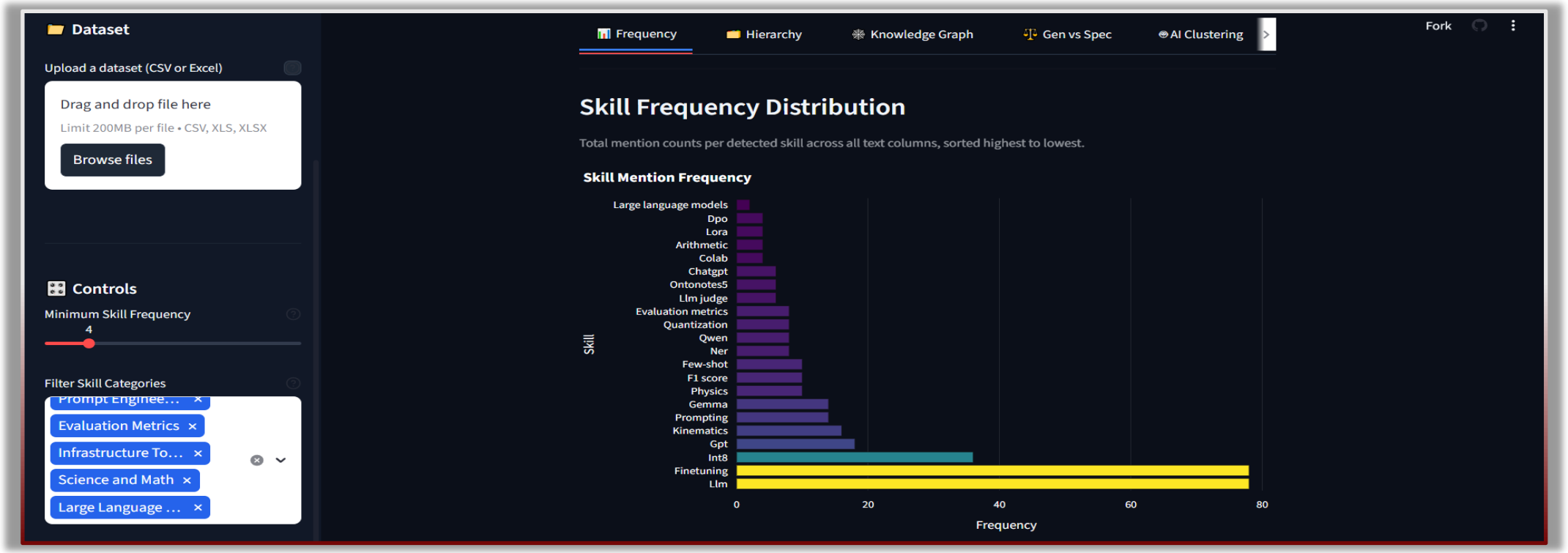
- Automatic skill extraction from logs
- Hierarchical skill categorization
- Interactive analytics dashboard

The system enables *data-driven understanding of team skills and expertise*.

System Demo

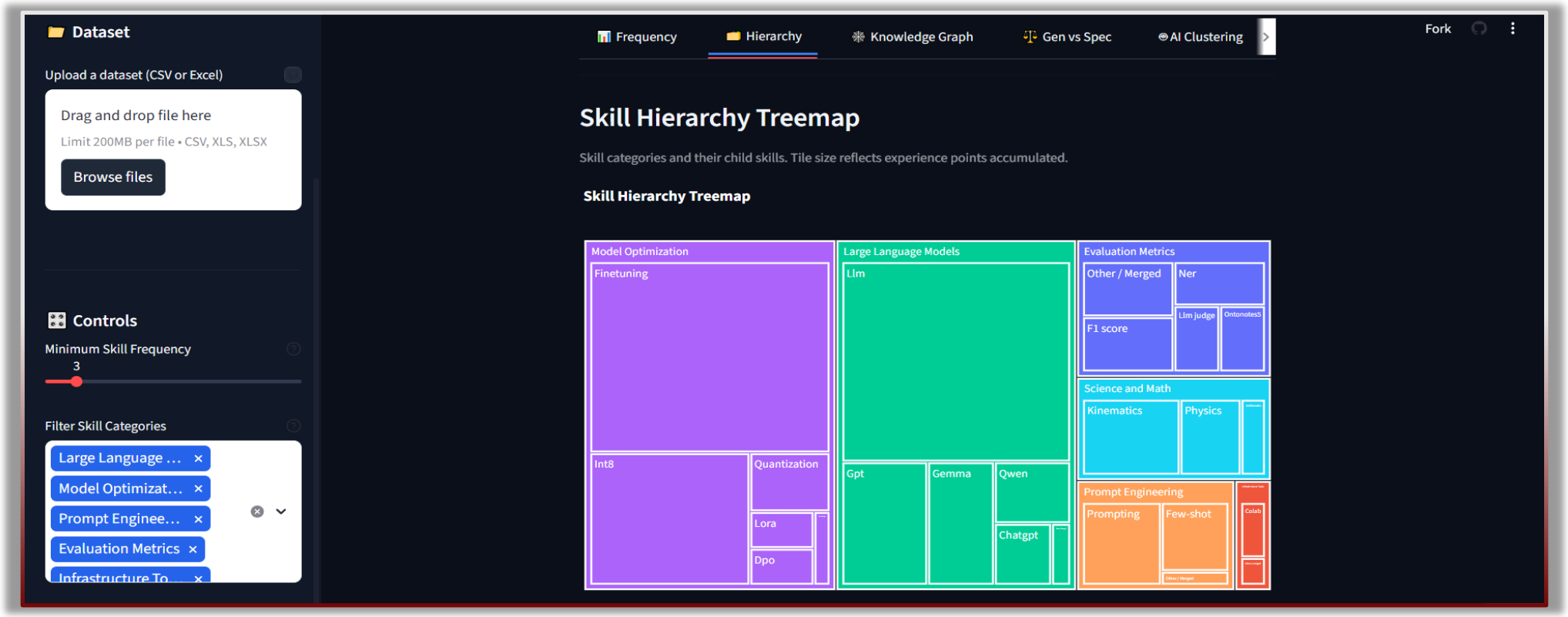

Skill Intelligence
Live Feed
Stream Live Logs
Dataset
Upload a dataset (CSV or Excel)
Drag and drop file here
Limit 200MB per file • CSV, XLS, XLSX
Browse files
Skill Intelligence Dashboard
Data source: activity_log.csv (default)
Total Experience Points
360
Skill Categories
6
Specific Skills (above ...
20
Most Frequent Skill
Finetuning
78 mentions
Most Active Category: Model Optimization
Avg XP per Category: 60.0 pts
Total Unique Skills Detected: 26

Skill Distribution



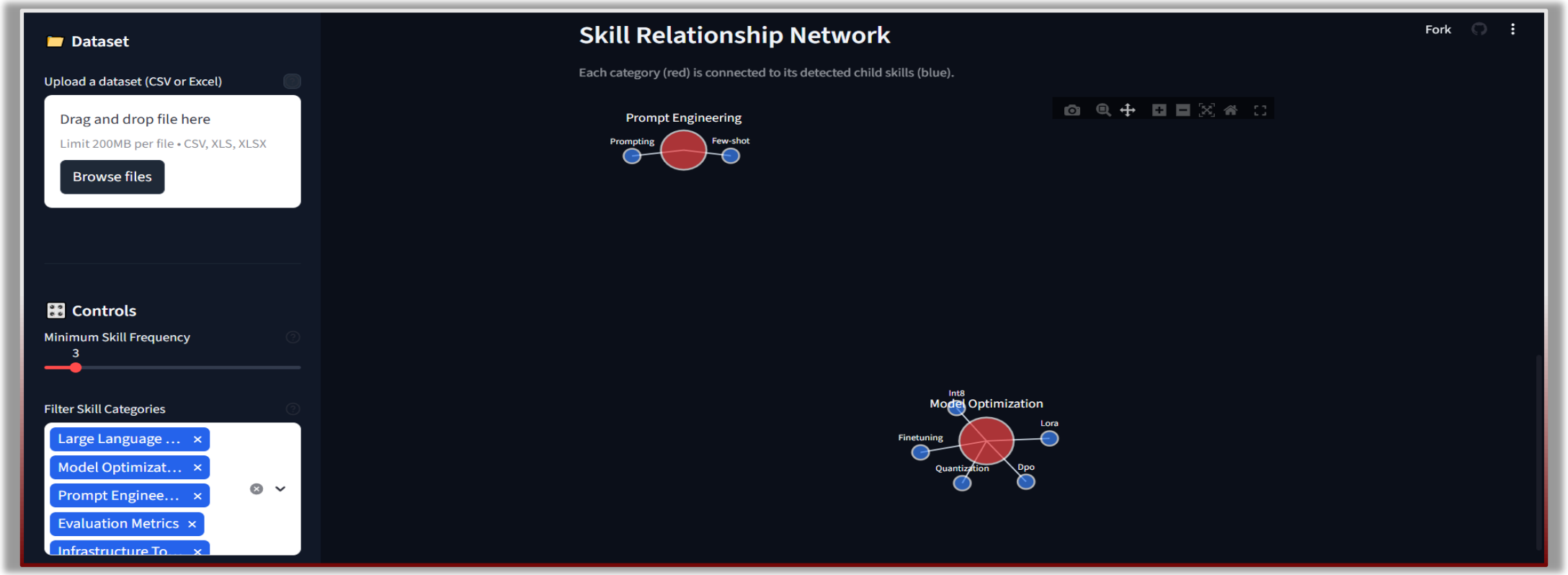
This chart shows the frequency of each detected skill across the dataset.

Skill Hierarchy



This visualization shows how specific skills belong to broader skill categories.

Skill Hierarchy



This graph shows *relationships between skills.*

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- Interactive analytics dashboard

The system enables *data-driven understanding of team skills and expertise*.

Links

GitHub Repository: https://github.com/kriti212/Skill-Intelligence-Dashboard-App
Demo: Streamlit Dashboard

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